

Project Initialization and Planning Phase

Date	2 July 2026
Team ID	
Project Title	Rising Waters – AI Based Flood Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to build an intelligent flood prediction system using machine learning to forecast flood risks based on weather parameters. It addresses the need for early warnings and efficient disaster management, enabling authorities to make timely decisions, save lives, and minimize property damage.

Project Overview	
Objective	The primary objective is to develop an accurate and reliable flood prediction system using machine learning techniques to provide early warnings and support disaster management.
Scope	The project collects, processes, and analyzes weather data such as rainfall, temperature, humidity, and water levels to predict flood occurrence and risk levels for specific regions.
Problem Statement	
Description	Lack of timely and accurate flood prediction leads to delayed warnings, inadequate preparedness, and significant loss of lives and property.
Impact	Solving this issue will enable early flood detection, improved disaster response, efficient resource allocation, and enhanced public safety.
Proposed Solution	
Approach	Utilizing machine learning algorithms to analyze historical and real-time weather data to predict flood risks and provide actionable insights.
Key Features	– Machine learning-based flood prediction model. – Real-time weather data integration. – Risk level classification (Low, Medium, High). – Early warning alerts and notifications. – Interactive dashboard for visualization and monitoring.

	- Real-time prediction and alerts for early flood warnings. - Continuous learning to adapt to changing weather patterns and improve prediction accuracy.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, Pycharm
Data		
Data	Source, size, format	Kaggle Flood Dataset, approx. 50,000 records, csv UCI Hydro Dataset, approx. 10,000 records, csv (Weather & Water Level Data)